

Exploring the repository of *de novo* designed bifunctional antimicrobial peptides through deep learning

Reviewed Preprint

v1 • May 28, 2024

Not revised

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Abstract

Antimicrobial peptides (AMPs) are attractive candidates to combat antibiotic resistance for their capability to target bio-membranes and restrict a wide range of pathogens. It is a daunting challenge to discover novel AMPs due to their sparse distributions in a vast peptide universe, especially for peptides that demonstrate potencies for both bacterial membranes and viral envelopes. Here we establish a *de novo* AMP design framework by bridging a deep generative module and a graph-encoding activity regressor. The generative module learns hidden ‘grammars’ of AMP features and produces candidates sequentially pass antimicrobial predictor and antiviral classifiers. We discover three bifunctional AMPs and experimentally validated their abilities to inhibit a spectrum of pathogens *in vitro* and in animal models. Notably, P076 is a highly potent bactericide with the minimal inhibitory concentration of 0.21 μ M against multidrug-resistant *A. baumannii*, while P002 broadly inhibits five enveloped viruses. Our study provides feasible means to uncover sequences that simultaneously encode antimicrobial and antiviral activities, thus bolstering the function spectra of AMPs to combat a wide range of drug-resistant infections.

eLife assessment

This study presents a **useful** pipeline for de novo design of antimicrobial peptides active both against bacteria and viruses. The method is based on deep learning, using a GAN generator and a regression tasked to predict antimicrobial activity. The evidence supporting the conclusions is promising but **incomplete**: three generated peptides are studied experimentally in vitro, and one is then tested in vivo in mice; the comparisons to other design methods could also be strengthened. This work will be of interest to the community working on machine learning for biomedical applications and specifically on antimicrobial peptides.

<https://doi.org/10.7554/eLife.97330.1.sa3>

Introduction

In the post-antibiotic era, microbes rapidly evolve to become antimicrobial-resistant under the pressure of antibiotic abuse and through horizontal transfer of resistance genes.¹ Due to increasing cases of untreatable bacterial infections, the World Health Organization (WHO) has projected a looming health threat of 10 million casualties per year by 2050.² Specifically, the WHO has identified six antimicrobial-resistant pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species) as ESKAPE. The ESKAPE pathogens have severely limited the treatment options for serious infections and impaired the conditions of immunodeficient patients.³ In sharp contrast, the majority of clinically available antibiotics were developed more than 30 years ago.⁴ Hence, there are urgent needs for targeted generation and screening of novel molecules with desired therapeutic properties for the next-generation broad-spectrum antimicrobial agents.

As a naturally evolved defense line of innate immune system, antimicrobial peptides (AMPs) are cationic macromolecules (usually 10-60 amino acids) widely present in all living organisms.⁵ AMPs are attractive alternatives to conventional antibiotic treatment because they effectively inhibit a wide range of pathogens, such as the Gram-positive and Gram-negative bacteria, fungi, and viruses.⁵ The majorities of AMPs featured amphiphilic patterns of amino acid sequences and enriched positive charges (e.g., lysine and arginine residues), which allowed AMPs to disrupt the membrane components and exert antimicrobial effects.⁶ Due to the relative constant compositions of biological membranes as the essential AMP targets, development of resistance to AMPs were rare events even after 100 passages of lab evolutions.^{7, 8}

Notably, the naturally evolved and rationally designed AMP sequences (~20 000) are housed in several databases (APD,⁹ DBAASP,¹⁰ DRAMP,¹¹ etc.) and paired with experimentally validated activities (minimal inhibitory concentration, or MIC), thus facilitating data mining endeavors for more diverse sequences and more effective peptides.¹² Indeed, deep learning models have led to the discovery of highly efficient AMPs (e.g., MIC of 2 μ M against *E. coli*). Specifically, predictive models like Deep-AMPEP30,¹³ AMPLify,¹⁴ TransImbAMP,¹⁵ and sAMPpred-GAT¹⁶ utilized deep convolutional, recurrent, attention-based, and graph neural networks to learn and recognize latent representations of meaningful peptides. These models recently enabled comprehensive mining of the human gut microbiome,¹⁷ modern and ancient human proteomes,^{18, 19} or even the entire sequence space of hexapeptides.⁸ Generative models like PepVAE,²⁰ CLaSS,²¹ and HydrAMP²² reconstructed the sequences from latent

space according to the encoder-decoder architecture of variational autoencoders (VAE), while PepGAN²³ and AMPGAN v2²⁴ utilized the generative adversarial network (GAN) to approximate the antimicrobial sequences by deceiving the discriminator.

Despite the recent advances of data-driven AMP classifiers and generators, the exploration of a vast AMPs repertoire is still hindered by several factors: (1) The sparsity of AMPs in the whole peptide space, which demands accurate prediction of MIC values for efficient sampling and filtering of AMP analogues.^{22, 25} The current machine learning predictive models usually perform the binary classification task and label the generated sequences as active or not, instead of giving specific activity values. To overcome this shortcoming, a popular pipeline combines independent modules of filtering, classification, and regression, but it faces risks of step-wise bias accumulation.⁸ (2) The impact of different phospholipid species on the efficacy of AMPs. Due to the data incompleteness of antimicrobial measurements, a majority of AMP classifiers or MIC predictions were specific to *E. coli* (Gram-negative). Nevertheless, these models were applied to find peptides that inhibit other Gram-positive bacteria (e.g., *S. aureus*), based on the assumption that the shared membrane components are potential targets of AMPs. More importantly, the enveloped virus also possesses a targetable membrane, which closely mimic the phospholipid components of cellular organelles. Even though the AMP database recorded antiviral activities of at least 2 000 peptides (AVPs), whether the current models can readily generate antiviral peptides remain elusive. (3) Lack of rigorous pre-clinical models. To advance the designed AMPs towards clinical trials, the experiments should reflect the efficacy of AMPs in tissue samples and require the determination of safe doses of a particular sequence before practical applications. To our knowledge, more than 120 AMPs have been suggested by deep learning models in the past three years, only a fraction of these studies demonstrated animal results.

In the current study, we addressed these shortcomings and presented a new deep learning-based *de novo* AMP design framework, consisting of a GAN generator and an antimicrobial activity regressor named AMPredictor. The graph convolution network (GCN) of AMPredictor allowed us to effectively train a regression model for the task of MIC predictions. Through in-depth investigations of generated sequences (e.g., antiviral possibilities), we selected three top-ranking sequences (P001, P002, and P076) and evaluated their efficacies against different pathogens (four drug-resistant bacteria and five enveloped virus). Out of the three candidates, we found P076 is a potent bactericide with MIC of 0.21 μ M against multidrug-resistant bacterium. It also outperformed a clinically relevant antibiotic polymyxin B in terms of safety and efficacy, as evaluated in mouse models. Another peptide P002 broadly neutralized viruses in cellular infection models. We challenged our computational models to generate bifunctional AMPs and our results demonstrated that these peptides inhibited a spectrum of pathogens, but with delicate preferences towards bacterial or viral species.

Results

Bridging the generator and regressor for *de novo* AMP design

The overall architecture for designing functional peptides fuses a generative adversarial network and an antimicrobial activity predictor (AMPredictor, **Figure 1a**). We aim to explore beyond the known antimicrobial templates or accessible proteomics space. We trained the GAN module with 3 280 AMP sequences to capture the pattern of antimicrobial peptides through few-shot learning. The training goal of GAN is to reach a relative balance between the generator and discriminator (see Methods **Equation 1**). The generator takes random noise as input to produce fake samples, which are fed into the discriminator along with the real data. The discriminator tries to distinguish between the real and fake data, then returns a probability to update the weights of the networks.²⁶ Here, we employed a low-dimensional vector called amino acid factors (AAF) to encode the peptide sequences and describe their physicochemical features.²⁷ We monitored the

generated sequences at different training stages (epoch 1, 100, 200, 400, 800) and visualized the optimization process using two common dimension reduction approaches, uniform manifold approximation and projection (UMAP) and t-distributed stochastic neighbor embedding (t-SNE). We clearly observed the distributions of newly sampled sequences among the existing AMPs after mapping peptide AAF features onto the two-dimensional space (**Figure 1b**). With iterating and updating, the generator learns the feature distribution of actual data and generates diverse sequences.

We checked the novelty and diversity of generated sequences at epoch 1 000 by aligning them with our training set. The sequence similarities were evaluated by global alignment using BLOSUM62 matrix. We found the identity of each generated sequence was around 0.35 to the known database (**Figure 1c**), indicating a low level of similarity to the existing AMPs. With distinct sequence compositions, generated peptides still hold similar physiochemical features to real AMPs, as evidenced by six descriptors of aliphatic index, aromaticity, net charge, the ratio of hydrophobic amino acids, instability index, and isoelectric point (**Figure 1d**).²⁸ The distribution of these descriptors closely resembled the physiochemical features of AMPs in the training set. Hence, adopting residue features as the input of GAN generator can capture the functional properties of actual AMPs.

We then developed an independent module AMPredictor to estimate the MIC values of arbitrary peptide sequences. AMPredictor is a GCN-based regression model, equivalent to the ‘score function’ in virtual drug screen. Unlike the low-dimensional descriptor adopted by GAN generator, we designed distinct representations of peptides in AMPredictor for model accuracy and orthogonal verification of GAN-generated sequences. We chose graph as the representation of peptides, with their amino acid residues as graph nodes and the predicted contact maps as graph edges. The node features were embeddings from the protein language model ESM.²⁹ The peptide graphs were then fed into a three-layer GCN to learn the hidden ‘grammars’ of AMP features. In addition to the sequence-level and structural-level features, we transformed the input amino acid sequences into Morgan fingerprints to encode the chemical features at each residual position. This atomic-level information was concatenated with GCN representations before entering a three-layer fully-connected networks to regress on MIC datasets (**Figure 1a**). AMPredictor obtained a low root mean squared error (RMSE) of 0.5348 and a high Pearson correlation coefficient (PCC) of 0.7072 on the test set (10% size of the dataset) (**Figure 1b**). The output of AMPredictor is the logMIC value of an AMP sequence.

By integrating the representations of residual-level, sequence-level, and protein structural level information, AMPredictor obtained the best performance in comparison to its ablating versions (Table S1). The ablation study indicated that the context embeddings from ESM play vital roles in predicting MICs, and the contact map assisted in improvement despite it being a simplified representation of protein structure. We also compared the performance of AMPredictor with five typical deep neural networks as baselines, including the multilayer perceptron (MLP), convolutional neural network (CNN), gated recurrent unit (GRU), long short-term memory (LSTM), and Transformer. AMPredictor outperforms these methods in terms of accuracy at the task of predicting antimicrobial activity values (scatter plots in Figure S2a). Hence, a well-trained generator could produce a variety of AMP candidates and then AMPredictor ranked the candidates by MIC values. Together with coarse to fine representations of AMP samples, the GAN and AMPredictor facilitated *de novo* design and *in silico* evaluation of AMPs.

Designing bifunctional antimicrobial peptides *in silico*

We acquired 104 novel peptides via the *de novo* design framework. Among them, 42 and 26 were predicted to own MIC values below 10 μ M and 5 μ M, respectively (supplemental spread sheet). To establish a sequential workflow and filter out antimicrobial peptides with antiviral capabilities, we then considered assembling several AVP classifiers subsequent to the AMPredictor. As the source data, the AVPdb provided ~2 000 sequences with IC₅₀ labels.³⁰ However, data curation

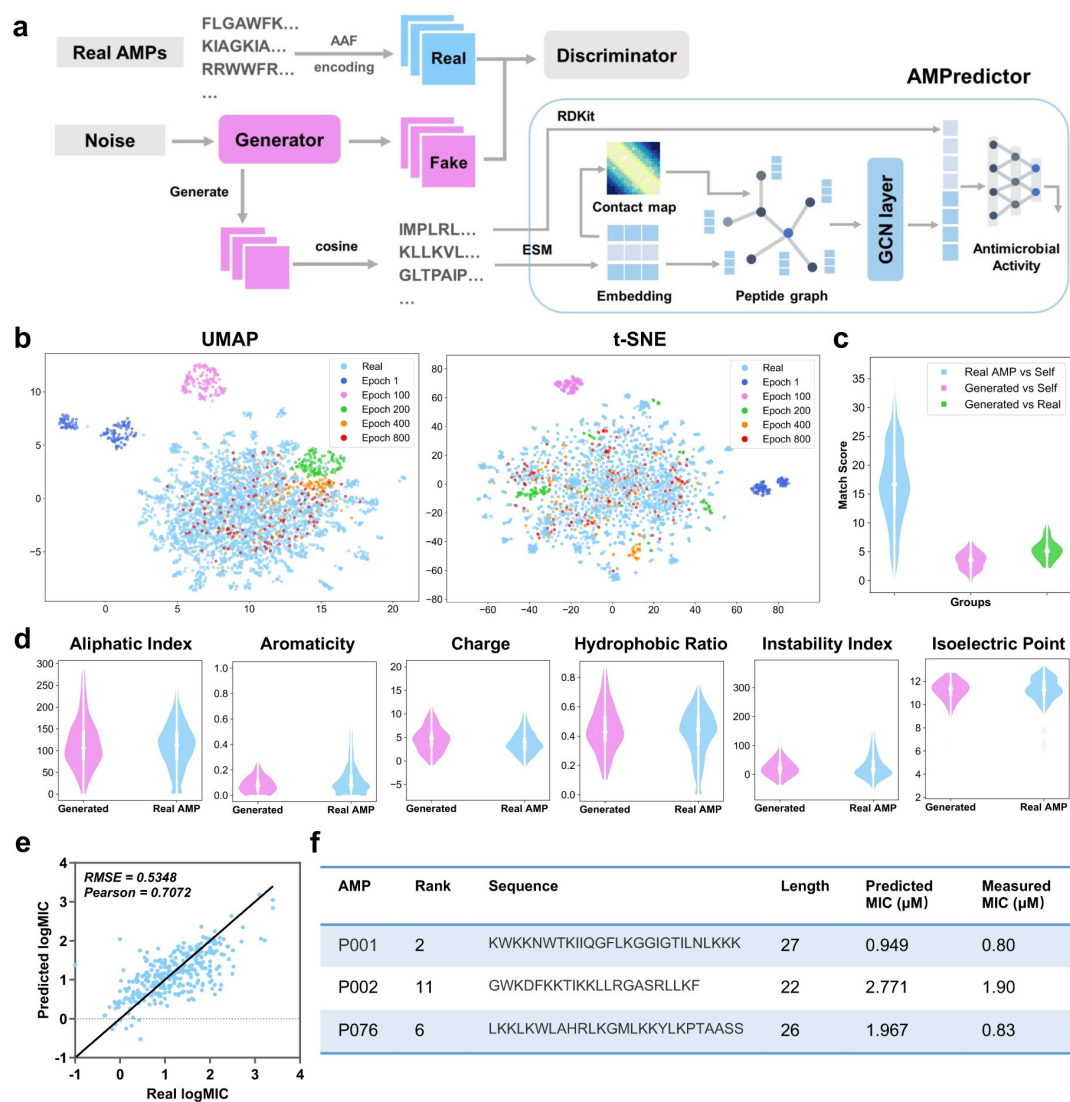


Figure 1.

Deep learning-based design of novel antimicrobial peptides.

a. The overall framework of GAN generator and AMPredictor.

b. Visualization of GAN training process via UMAP and t-SNE. The newly generated sequences fit the distribution of known AMP space gradually.

c. Global alignment match scores between the generated sequences and itself and AMPs in the training set.

d. Six physiochemistry properties of generated peptides and real AMP sets.

e. Regression results of AMPredictor model on its test set.

f. The sequences of three selected peptides as well as their predicted and experimentally validated MIC values against *E. coli* ATCC 25922.

resulted in only 759 unique AVPs. We determined that the quantitative data of AVPs were insufficient for training another GCN-based regression model. Instead, binary classification should exhibit robustness against noise of a relatively small dataset. Hence, we merely made yes/no decisions using a collective of five published AVP classification tools (AVPpred,³¹ DeepAVP,³² Deep-AVPpred,³³ ENNAVIA,³⁴ and AI4AVP,³⁵ details in Table S2). We regarded a peptide as an antiviral candidate if four or more positive votes were recorded and 32 peptides met our criteria.

We conducted a comprehensive evaluation of efficacy and safety, to select novel AMPs before experimental validations. Considering the membrane corruption mechanism of most AMPs, we predicted their hemolytic activity as a quick measure of cellular toxicity.³⁶ Through the ranking of high activity and low toxicity, we picked three potent peptides P001, P002, and P076 (**Figure 1f**), all of which featured amphiphilic signatures of sequence and enriched cationic regions (e.g., helical wheel projections in Figure S3), for *in vitro* and *in vivo* experimental assessment.

***In vitro* antibacterial and hemolytic assays**

We examined the antibacterial activity of P001, P002, and P076 against one Gram-positive (*S. aureus*) and three Gram-negative (*A. baumannii*, *E. coli*, and *P. aeruginosa*) bacteria, including three ESKAPE strains. For each species, we cultured a standard strain and a multidrug-resistant (MDR) strain, and measured the MIC values of individual peptides using the broth microdilution method. All the peptides exhibited inspiring bactericidal properties with MICs ranging from 0.20 to 15.18 μM (corresponding to 0.625 to 40 $\mu\text{g/mL}$, **Figure 2a**). P001 showed broad-spectrum antibacterial capability with MICs less than 4 μM across all the eight strains of four bacteria species. P002 featured lower MIC values in inhibiting Gram-positive bacteria *S. aureus*, while P076 mainly inhibits Gram-negative bacteria *A. baumannii*, *E. coli*, and *P. aeruginosa*. Notably, the experimentally measured MICs against *E. coli* agreed well with the predictions of AMPredictor (**Figure 1f**), indicating the accuracy of our regression models.

The hemolytic analysis revealed that both P001 and P002 displayed dose-dependent toxicity to mouse erythrocytes (**Figure 2b**), with average hemolysis higher than 50% at high concentrations (20 μM , or 100 $\mu\text{g/mL}$). Nevertheless, the hemolysis of P076 was lower than 1% at concentrations up to 70 μM . We also inspected the apparent cytotoxicity (CC_{50}) of three peptides on three mammalian cell lines (Vero-E6, A549, Huh-7) using CCK8 assays. For the cell cultures, CC_{50} values of all three peptides were well above 30 μM (Figure S8a). The CC_{50} of P076, specifically, were in the range of 35.5 μM (for A549 cells from lung tissues) to 66.8 μM (for Huh-7 cells from liver tissues). The *in vitro* safety assays pointed out thresholds of concentration for administrating these peptides, which were adapted in subsequent animal experiments for estimation of safe dosage.

Our data suggest that all three peptides are potential candidates for inhibiting the clinically relevant *A. baumannii*, which is a top-ranking pathogen with critical priority for R&D of new antibiotics, as defined by WHO.³⁷ The MIC of P076 at eliminating the persistent MDAB reached 0.21 μM , a value half of P001 and P002 (0.40 and 0.47 μM , respectively). Meanwhile, P076 was comparable with P001 but more efficient than P002 against *E. coli* and *P. aeruginosa*. Despite a slight inferiority in killing *S. aureus*, P076 owned a better biosafety. Taken together, the sequence of P076 likely leveraged the balance between efficacy and safety, we therefore focused the following *in vivo* evaluation and animal infection models on P076.

***In vivo* antibacterial and toxicological evaluations**

We first determined a safe dosage for administrating P076 on mouse models. A total of 140 C57 mice (randomly divided into 14 groups) were given increasing concentrations of P076 through intraperitoneal injection, and the cyclic peptide antibiotic polymyxin B (PB) was employed as a positive control. In spite of nephrotoxicity, PB is one of the polymyxin antibiotics that is considered

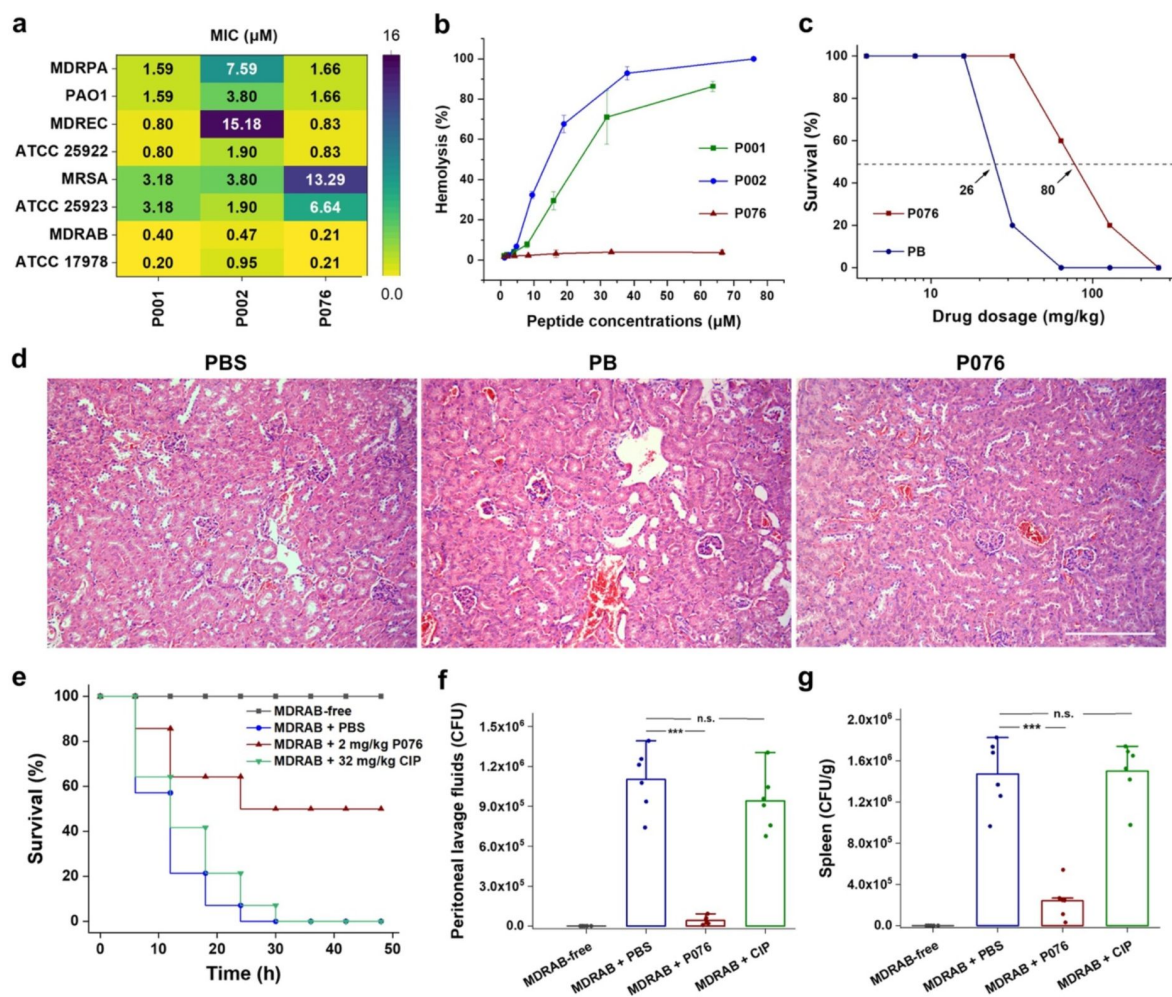


Figure 2.

Antibacterial and toxicological evaluations of P076 peptide.

a. *In vitro* antibacterial assessment of peptides by determination of MICs.

b. Hemolysis of increasing concentrations of peptides. Results are shown as the means $\pm$ standard deviation (SDs).

c. Survival curves of mice given P076 and PB. Arrows indicate the drug concentration inducing a 50% death.

d. H&E staining of mouse kidneys. The scale bar indicates 200 μm .

e. Survival of infected mice treated with P076 and CIP.

f. g. Bacterial colonization of MDRAB in mouse peritoneal lavage fluid and spleen. Results are shown as the means $\pm$ SDs. ***, $P < 0.001$. n.s., no significance.

“the last ditch” for the treatment of infections caused by extensively drug-resistant Gram-negative bacteria in clinic.³⁷ Animal survival was monitored for 48 h after a single drug administration. The P076 concentration to induce a 50% death was calculated to be 80 mg/kg, approximately 3-fold higher than that of PB (26 mg/kg, **Figure 2c**). Hematoxylin-eosin (HE) staining further verified that no pathological changes were observed in the mice liver, spleen and kidneys of mice given 32 mg/kg P076 after 12 h, whereas an apparent renal tubular injury was discovered in PB-treated mouse (**Figure 2c** and S5a).

Encouraged by the biocompatibility of P076, we further conducted an *in vivo* antibacterial assay using a mouse peritoneal infection model.³⁸ A total of 3×10^7 CFU of MDAB was intraperitoneally injected in C57 mouse to establish infection symptoms. P076 was administered at 2 mg/kg at 0.5 and 2.5 h after bacterial invasion. As the positive control treatment, we employed a single injection of 32 mg/kg ciprofloxacin (CIP) at 0.5 h after infection. The mice given PBS (blank buffer) or CIP all deceased after 48 h, while 50% of the mice treated with P076 remained alive (**Figure 2e**). Additional investigations revealed that the bacterial colonization in the peritoneal lavage fluid of PBS-treated mouse was averagely 1.1×10^6 CFU, which was 25 times higher than that in the mice treated with P076 (4.2×10^4 CFU, $P < 0.001$), whereas CIP featured less of effect on the number of bacteria (**Figure 2f**). Similarly, P076 dramatically decreased the bacterial loads in the mouse spleen (about seven times lower, **Figure 2g** and S5b), thus demonstrating the prospect of P076 to be developed into a new antibiotic.

Broad-spectrum antiviral assays

A cohort of traditional antimicrobial peptides have been rediscovered with antiviral abilities.³⁹ One significant reason is that their membrane-disrupting capacity can adapt to both bacterial membranes and viral envelopes, which may be captured by AMPredictor and AVP classifier filters. Thus, we tested the antiviral activities of P001, P002, and P076, which passed our dual filters. Our experiments included five typical enveloped viruses. Chikungunya virus (CHIKV), Hantaan virus (HTNV), Dengue virus 2 (DENV-2), Herpes simplex virus 1 (HSV-1), and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2, and the BA.2. omicron variant) are all pathogens of the past or ongoing epidemics, such as the chikungunya fever, hemorrhagic fever with renal syndrome, dengue, encephalitis, and COVID-19. Hence, the peptide drugs may be repurposed as preventions against viral infections.

We incubated the peptides with different viruses before they invaded the cultured cells and self-amplified. We analyzed the potential antiviral effects of P001, P002, P076 at the mRNA and protein levels, respectively. At the mRNA level, the quantitative real-time PCR suggested dose-dependent inhibition of viral-specific gene expressions (E1 gene for CHIKV, S gene for HTNV, VP16 gene for HSV-1, NS5a gene for DENV-2, ORF for SARS-CoV-2, **Figure 3a** and **Figure S6b**). At the protein level, we applied fluorescence-conjugated antibodies to visualize trace amounts of virus populations. Corroborating with the RT-PCR results, P002 nearly abolished all viral infections at 12.5 μ M (**Figure 3b**, **Figure S6a** and **Figure S7**). To quantify the immunofluorescence results, we plotted the log-scale fluorescent focus forming unit (FFU) against varying peptide concentrations (**Figure S8b**). Among the three peptides, P002 exhibited the highest potential with EC_{50} of 0.37 μ M (against CHIKV) to 2.08 μ M (against HTNV) (**Figure 3b**). The EC_{50} of P076 and P001 were in the range of 1.62 to 2.67 μ M (except for inhibiting DENV-2).

Combining the EC_{50} and CC_{50} calculations, we defined a selectivity index ($SI = CC_{50}/EC_{50}$) as the experimental guides of safety versus efficacy (Table S3). The SI of P002 against four viruses are 149.21, 31.75, 51.05, and 59.01, three of which outperformed the SI of P001 and P076, especially for the case of HTNV. Hence, we further observed the performance of P002 with transmission electron microscopy. For the control set, the viruses were allowed to propagate in cells for days and identified as assembled particles in cytosol (**Figure 4a**, b). When the viruses were mixed with 10 μ M P002 before addition to the cells, the viral particles became absent while the cell morphology kept intact, indicating the broad-spectrum inhibition effects of P002 (**Figure 4c**, d).

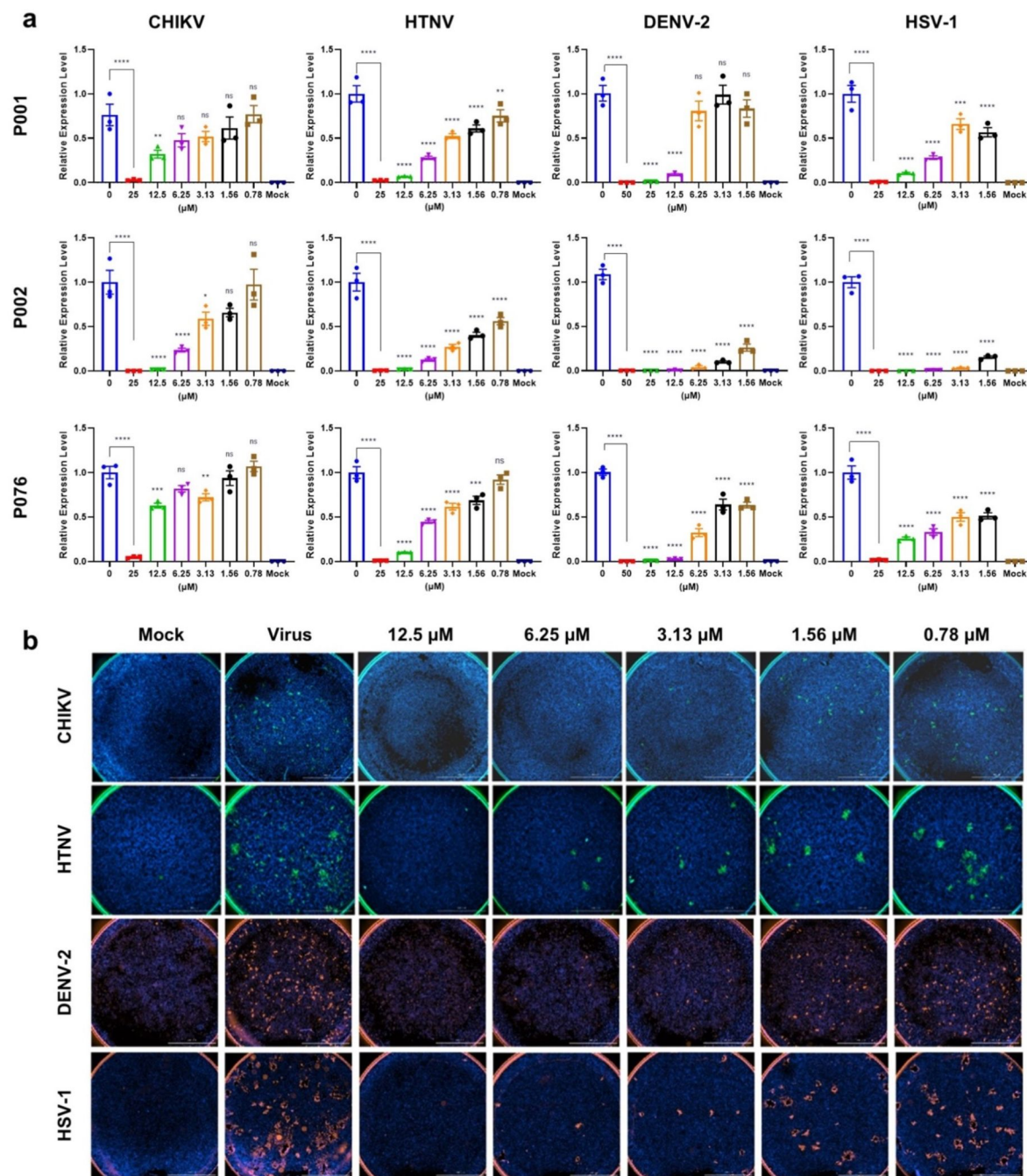


Figure 3.

Antiviral assays of AMPs against CHIKV, HTNV, DENV-2, and HSV-1.

a. Quantitative real-time PCR of virus RNA with gradient concentrations of P001, P002, and P076.

b. Immunofluorescence results of four viruses with gradient concentrations of P002.

All experiments are performed in triplicates. Column bars are means $\pm$ SEMs. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. ns, no significance.

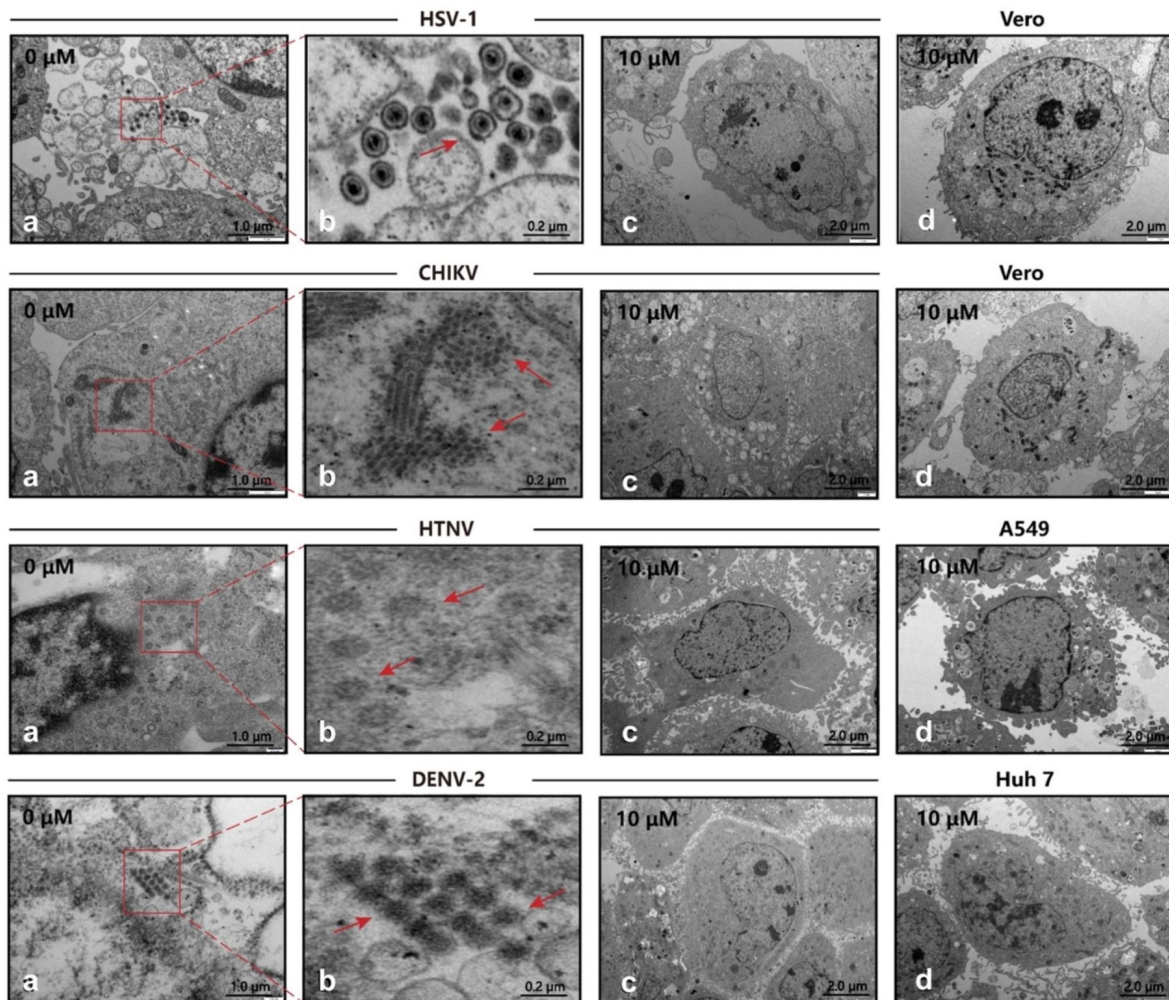


Figure 4.

Transmission electron microscopy (TEM) of viral infection with P002.

a. b. Cells infected with virus, where the red arrows mark the viral particles.

c. Cells are treated with 10 μ M P002 and infected with virus (MOI=1), where no viral proliferation is observed.

d. Control of intact cells with 10 μ M P002. All experiments are performed in triplicates.

Because we observed a gradient of antimicrobial and antiviral activities from P001, P002, and P076 with their individual preferences against bacteria and viruses, we aimed to probe the interactions between peptides and different phospholipid assemblies, thus deriving feasible means judging their underlying preferences. We performed all-atom molecular dynamics simulations with molecular models of bacterial membranes and viral envelopes, placed each peptide adjacent to the phospholipid bilayers, and simulated them attaching to the membrane (Figure S9a, b). For the model of Gram-negative bacteria systems, the MM/PBSA calculations and ΔG_{bind} distinguished P002 from P076 and P001 (Table S8). Evaluated by ΔG_{bind} (-465.79 ± 11.39 kJ/mol), P076 would associate with the membrane of Gram-negative bacteria tighter than others. For the model of viral envelopes, all binding energies were weaker than that to bacteria. The ranking of $\Delta G_{\text{nonpolar}}$ corroborated with the antiviral potential of P002. Combining this energy term with a geometrical parameter (average distance between the COM of peptide and phosphorus atoms in the top leaflet on z-axis) may predict the behavior of P002 better than $\Delta G_{\text{nonpolar}}$ alone (Figure S9c).

Membrane-penetration mechanisms

We hypothesized that a common target among bacteria and viruses was the membrane components. To shed light on the possible membrane-penetration abilities of designed peptides, we applied quantitative measurements, first by focusing on signature components of Gram-negative bacterial outer membranes. We measured the binding of three peptides to Lipopolysaccharide (LPS) via biolayer interferometry (BLI). More P076 and P001 peptides were recruited to immobilized LPS than that of P002 at 500 nM (Figure 5b). Further experiments demonstrated that P076 bound to lipid A, the innermost part of LPS, at an equilibrium dissociation constant (K_D) of 56.1 nM. This binding affinity of P076 to lipid A was slightly higher than that of P001 (65.7 nM) and was approximately 2.4-fold less than that of P002 (135 nM, Figure 5a). Then we directly investigated the membrane permeability by 1-N-phenyl naphthylamine (NPN) uptake assay. NPN is a fluorescent dye for detecting the impaired outer membrane of Gram-negative bacteria. P076 elicited a stronger membrane penetration relative to P001 and P002 (wavelength scanning in Figure 5c). Together, our results suggested that P076 could efficiently bind to and damage bacterial membranes.

To gain a more comprehensive understanding of AMP-membrane interactions with fine content of amphiphilic assemblies (both outer membranes and inner membranes), we also observed the membrane attachment and partial insertion of AMPs via all-atom molecular dynamics simulations. The top leaflet of Gram-negative bacterial outer membranes was built as simplified *E. coli* LPS (lipid A and R1 core). With an initial equilibrated position above the membrane surface, the peptide started to approach the oligosaccharides of LPS and then bind to them. During the simulation processes, P002 and P076 gradually inserted into the outer leaflet, and represented poses binding to LPS at 100 ns (Figure 5e). We also measured the distance between the COM of peptide and phosphorus atoms of lipid A on z-axis in the latter 50 ns of trajectories (Figure 5d). Compared with the initial value about 4 nm (showing as dashed line), this metric of P002 and P076 declined to less than 2 nm. Nevertheless, P001 performed weaker interactions with an increasing distance, which was consistent with the membrane-disrupting experiment in Figure 5c. Moreover, three AMPs tended to attach to the membrane in the MD simulations with lipid bilayer system mimicking the inner membranes of Gram-negative bacteria (Figure S9b). Collectively, experimental characterizations along with computational simulations indicate the membrane-acting mechanisms of our AMPs when playing antimicrobial roles. Further explanation of membrane-disrupting details or other potential biomolecular targets remains to be explored in the future.

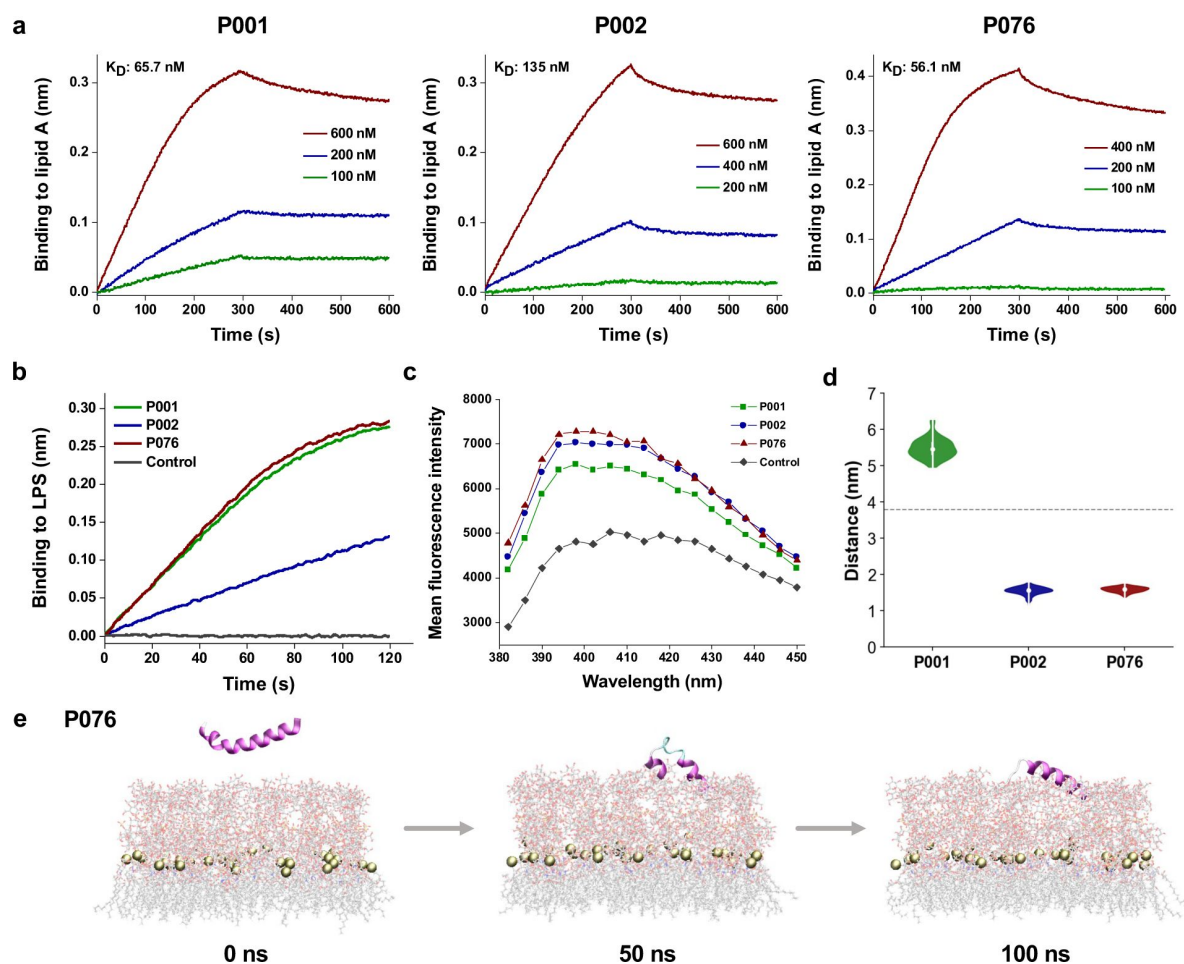


Figure 5.

Bacterial membrane-acting assessment of AMPs.

- Binding kinetics of three AMPs to lipid A measured by biolayer interferometry.
- Binding thicknesses of 500 nM AMPs to immobilized LPS within 120 seconds.
- Fluorescence intensity of NPN after excitation at 350 nm. The maximum emission appeared at approximately 396 nm, where P001, P002, and P076 were detected higher intensity than control.
- The distance between COM of peptides and phosphorus atoms of lipid A (around the midplane of G-bacterial outer membranes) during the last 50 ns of the MD simulation trajectories. The gray dashed line displays the initial distance at 0 ns.
- Snapshots of P076 with G-bacterial outer membranes at 0 ns, 50 ns, and 100 ns. Only the top leaflet (LPS) is shown. Tan balls represent the phosphorus atoms of lipid A molecules.

Discussion

The traditional ‘lock and key’ model for therapeutic molecule design requires knowledge of target structures.⁴⁰ When the targets are biological membranes, the means of docking and ranking by binding affinities (K_D) can easily become futile. The rapid development of deep neural networks, on the other hand, has facilitated self-supervised learning of meaningful molecule representations in hidden dimensions.²⁹ Combined with vast data of known AMPs, numerous deep learning models have enabled mining of new antibiotics.^{18, 19} In the current study, we deployed a new combination of descriptors to extract representations of biological sequences at residual and protein levels. First, we transformed input amino acid sequences into Morgan fingerprints to encode the chemical features of each residue. Second, the scaffolds of AMPs are closely relative to their antimicrobial function, but rarely considered at the structural level in recently reported models (except for sAMPpred-GAT¹⁶). Instead of using limited amounts of crystal structures, we introduced predicted contact maps of peptides as the edge encoding of peptide graphs. Finally, we added the embedding from ESM as the residue node features to present the associations of a single peptide sequence in the large protein repertoire. ESM was trained on millions of natural protein sequences via mask attention mechanism and assisted in few-shot learning via transferring the related knowledge of protein sequences.²⁹ The resulting AMPredictor accurately predicted the MICs of three *de novo* design AMPs in our study. As an independent test on 28 new AMPs discovered by different models, AMPredictor also scored well (supplementary Table S4).

Despite the ability of deep learning model to guide targeted generation and screen of functional peptides, it lacks the merit of molecular docking program to explicitly consider subtle environmental features. For a particular sequence, we found it was difficult to tell its preference towards bacterial membranes or viral envelopes, due to slight differences in their lipid compositions. Consequently, we observed broad antimicrobial activity for P001, broad antiviral activity for P002, while P076 demonstrated superior capacity on *A. baumannii*. As a control, we made a tiny change of P076 C-terminal with amide modification but observed loss of functions (Table S5). The change of a few atoms on an intact sequence can lead to dramatic functional consequences, which may be challenging for neural networks to capture.

Recent studies indicated unexpected side effects of traditional antibiotics. Instead of restraining the growth of carbapenem-resistant *Enterobacteriaceae* in the intestine environment, broad-spectrum antibiotics reduced gut microbial populations and depleted beneficial metabolites, thus facilitated colonization of *Enterobacteriaceae* species.⁴¹ With more limitations of conventional small molecule antibiotics discovered, AMPs are attractive alternatives to conventional antibiotic treatment. Their wide bioactivity spectra can expand to virus, thus minimizing the cost of screen multiple drugs. Our results suggested efficient *in silico* screen tools plus rigorous experimental validations could uncover highly potent peptide drugs without comprising on safety, and large-scale animal experiment could facilitate translation towards clinical research. In summary, we believe that a mature workflow involving sequence generation and evaluation can promote the discovery of ideal AMPs to eventually alleviate the hazard of drug-resistant diseases.

Methods

Data Curation

Our AMP dataset is collected from PepVAE,²⁰ containing 3280 antimicrobial peptide sequences with their minimal inhibitory concentrations (MICs) in logarithmic form. To integrate various AMP sequence and bioactivity data, Witten et al. developed the Giant Repository of AMP Activity

(GRAMPA).⁴² Since the MIC values change when targeting different microbial strains, only AMPs inhibiting *E. coli* were selected. Also, to simplify the input and reduce accidental effect, peptides without modifications are filtered so that no cysteine appears (amino acid composition in Figure S1a). The sequence length of AMPs are less than 40 amino acids, and the majority of lengths (~60%) are between 10 and 20 (Figure S1a). Labels, i.e., logMIC values, are distributed from -1 to 4. During the training process, the dataset is divided into the training, validation, and testing sets in a ratio of 8:1:1 randomly.

Generator

The generative adversarial network (GAN) is composed of a generator network and a discriminator. Through the co-training strategy, the generator evolves to produce ideal sequences from random noises z , fitting the distribution of real samples and faking the discriminator.

$$\min_G \max_D V(G, D) = E_{x \sim p_{real}(x)} [\log D(x)] + E_{z \sim p_{fake}(z)} [\log (1 - D(G(z)))] \quad (1)$$

For the combined training of GAN generator and discriminator, the 3280 AMP sequences from PepVAE were encoded with amino acid factors (AAF). AAF encoding represents five physiochemical values, including the polarity, secondary structures, molecular size, codon diversity, and charges.²⁷ The GAN architecture was utilized to generate novel AMP sequences, where three layers of 2D convolution were inside both the generator and discriminator.⁴³ Loss function modified with WGAN-GP strategy was taken to avoid mode collapse, and the optimizer was kept as RMSProp. The learning rate was 1e-4. Cross training ratio between generator and discriminator was set as 1:5. Deep learning-based models were trained on an NVIDIA GeForce RTX 2080 Ti GPU. After training for 1000 epochs, string sequences are calculated via cosine similarity. A batch of sequences were generated with the limited length of 30 amino acids. Redundant alanine (A) at the end of every sequence were removed, therefore leaving sequences with variable length.

Predictor

Antimicrobial peptide activity predictor (AMPredictor) was built based on PyTorch and PyTorch Geometric library. All input sequences were treated with padded zeros to reach unified lengths of 60. For chemical fingerprint, RDKit (www.rdkit.org) was used to transform each amino acid into 2048-bit Morgan fingerprint in peptide sequences. For each peptide, the fingerprint vectors were concatenated and contrasted into a 60-dimensional vector by average.⁴⁴ Pretrained language model esm1b_t33_650M_UR50S was downloaded and used to provide the embeddings and the contact maps.

During the training process, the optimizer was Adam, and loss function was the mean squared loss. Batch size and learning rate were decided upon grid-search, whose values were 128 and 0.001, respectively (Figure S1b). Evaluation metrics MSE, RMSE, PCC, and CI are calculated as following:

$$MSE = \frac{1}{n} \sum_{i=1}^n (b_i - a_i)^2 \quad (2)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (b_i - a_i)^2} \quad (3)$$

$$PCC = \frac{\sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_{i=1}^n (a_i - \bar{a})^2 \sum_{i=1}^n (b_i - \bar{b})^2}} \quad (4)$$

$$CI = \frac{1}{Z} \sum_{a_i > a_j} h(b_i - b_j) \quad (5)$$

where a_i is the real activity label of peptide i , and b_i is the predicted activity value, Z is the scale constant, and $h(b)$ is the step function.

For comparisons in Figure S2, five baseline models were implemented by DeepPurpose package.⁴⁵ The amino acid composition encoding was used as the input of MLP, while one hot encoding was for CNN, CNN-GRU, and CNN-LSTM. Fragment partition fingerprint of peptides was fed into the Transformer encoder.

In ablation study, three encoding approaches (chemical fingerprint, contact matrix, and ESM) and their different combinations were tested individually. It was noted that the peptide graph exists only when both the node embedding and contact matrix are present together. Therefore, a CNN with the same number of layers and output dimension was adopted instead of the GCN in ablation test.

Peptide synthesis

The peptide P001, P002, and P076 were synthesized by DGpeptide Co., Ltd. with 95% purity via solid-phase peptide synthesis procedures. Mass spectrometry were used to verify the molecular weights (Figure S4).

MIC detection

The minimum inhibitory concentrations (MICs) of peptides against bacteria including *A. baumannii* (ATCC 19606), multidrug-resistant *A. baumannii* (MDRAB),⁴⁶ *S. aureus* (ATCC 25923), methicillin-resistant *S. aureus* (MRSA, ATCC 43300), *E. coli* (ATCC 25922), multidrug-resistant *E. coli* (MDREC),⁴⁷ *P. aeruginosa* (PAO1), and multidrug-resistant *P. aeruginosa* (MDRPA),⁴⁸ were determined using the broth microdilution method recommended by the Clinical and Laboratory Standards Institute (CLSI) with modifications reported by the Hancock laboratory.⁴⁹ P001, P002, and P076 peptides were prepared in 0.01% acetic acid containing 0.2% BSA at concentrations of 6.25, 12.5, 25, 50, 100, 200, 400, 800, 1600, and 3200 $\mu\text{g/mL}$. Bacteria grown to the mid-logarithmic phase in Meller-Hinton broth (MHB) were diluted to 2×10^5 CFU/mL. Aliquot of 100 μL of the

bacterial suspension was then co-incubated with 11 μL of the peptides at 37°C in a sterile polypropylene microtiter plate (3365, Corning, Shanghai, China). An M2e microplate reader (Molecular Devices, Silicon Valley, CA, USA) was applied to detect the bacterial absorbance at 600 nm after 18 h. The recorded MIC was the lowest concentration of peptides causing a 70% reduction in bacterial turbidity.

Hemolytic assay

The hemolysis of peptides was evaluated as previously described in Ref.⁵⁰. Briefly, the fresh mouse blood was centrifuged at 1000 g for 10 min and washed with cold PBS three times. The erythrocytes were collected and diluted in saline solution to prepare a 2% (v/v) suspension. A total of 75 μL of the erythrocyte suspension was co-incubated with 75 μL of the peptides (6.25, 12.5, 25, 50, 100, 200, and 400 $\mu\text{g}/\text{mL}$) at 37°C for 1 h. After centrifuging at 1000 g for 10 min, 60 μL supernatant was obtained, and the absorbance was determined at 414 nm. This experiment was conducted in triplicate and repeated twice.

In vivo antibacterial evaluation

The therapeutic efficacy of P076 peptide against drug-resistant bacteria was analyzed using a mouse peritoneal infection model.³⁸ The 8-week-old female C57 mice ($n = 10$) was intraperitoneally administered with 3×10^5 CFU/mL of MDRAW. P076 (2 mg/kg) was given by intraperitoneal injection at 0.5 and 2.5 h after bacterial infection. Ciprofloxacin (CIP, C129896, Aladdin, 32 mg/kg) was employed as the control. Mouse survival was monitored for 48 h. For detection of the bacterial colonization, another 24 mice were randomly divided into four groups ($n = 6$ for each group). The mice were euthanized 12 h after MDRAW infection, and the peritoneal cavity was rinsed with 2 mL of PBS. Additionally, mouse liver and spleen were obtained by surgery and homogenized in 1 mL of PBS. The lavage fluid and tissue suspension were both diluted for bacterial colony counting using an MHB agar plate as described in MIC detection.

In vivo toxicological evaluation

The 8-week-old female mice (18–22 g) were randomly divided into 14 groups ($n = 10$ for each group). P076 peptide and polymyxin B (PB, P105490, Aladdin, Shanghai, China) were given to the mice by intraperitoneal injection at 4, 8, 16, 32, 64, and 128 mg/kg. Mouse survival was monitored for 48 h. For pathologic analysis, mouse liver, spleen, and kidneys were obtained by surgery for hematoxylin-eosin (HE) staining after treatment with 64 mg/kg P076 and PB for 12 h. Mice were cared for and treated in accordance with the NIH guidelines for the care and use of laboratory animals (NIH Publication No. 85e23 Rev. 1985). Animal experiments were conducted with approval from the Animal Experimental Ethics Committee of the Army Medical University (AMUWEC20223290).

Cell cultures

Vero-E6 (African green monkey kidney cell line), Huh-7 (Hepatocellular carcinoma cell line), A549 (Human Lung carcinoma cell line) and HeLa-hACE2 were cultured in DMEM (Thermo Fisher Scientific, Waltham, USA) containing 10% fetal bovine serum in a 5% CO_2 incubator at 37°C. HSV-1 and CHIKV were propagated in Vero-E6 cells. DENV-2 and HTNV were propagated in Huh-7 and A549 cells, respectively. And SARS-CoV-2 were propagated in HeLa-hACE2 cells.

Cytotoxicity measurements

The Vero-E6, Huh-7 and A549 cells (2×10^4 cells/well) were inoculated in the 96-well culture plate and grown overnight to 90% confluence at 37°C. The cells were treated with different concentrations of peptides (P001, P002, and P076) for 1 hour, and the same concentration of PBS

was used as a control. After 48 hours of cell culture, 10% CCK8 solution was added to each well and incubated at 37°C for 1 hour. The absorbance was measured at 450 nm using a microplate reader. The CC_{50} represents the peptide concentration at the 50% of viability of uninfected cells.

Immunofluorescence assay

The cells were cultured in a 96-well plate (2×10^4 cells/well). Different viruses were mixed with P001, P002, or P076 peptides for 10 min before infecting the cells for 1 h. The infected cells were then cultured with DMEM containing 10% serum for 24–48 h. After discarding the supernatant and washing it three times with PBS, it was permeated with PBS containing 0.2% Triton X-100 (containing 0.1% BSA) at room temperature for 30 minutes. Then, the cells were mixed with an antiviral antibody at 4°C overnight. After washing with PBS three times, the cells were incubated with fluorescence-conjugated secondary antibodies at room temperature for 1 hour. Cell nuclei were stained using 4',6'-diamidino-2-phenylindole (DAPI) in the dark for 5 min, and images were obtained by fluorescence microscope.

Real-time PCR

The cells were cultured in a 6-well plate (2×10^5 cells/well) following the same cell culture procedures as in the immunofluorescence assay. Cellular RNAs were extracted and purified using a viral RNA extraction kit (Qiagen, Dusseldorf, Germany), and then cDNA was synthesized and monitored with SuperScript-III kit (Takara, Dalian, China). qRT-PCR analysis was performed using SYBR Green (BIO-RAD, California, USA) according to the manufacturer's protocol. Viral RNA expression was calculated using the $2^{-\Delta\Delta C_T}$ (cycle threshold) method normalized to GAPDH expression. The qRT-PCR primers sequences for HTNV, CHIKV, HSV-1 and DENV-2 are listed in the Table S6, while the detection kits (Daan gene, Guangzhou, China) were used for SARS-CoV-2 wild-type and BA.2 subtype.

Transmission electron microscopy

The cells were cultured in a 6-well plate (2×10^5 cells/well) following the same cell culture procedures as in the immunofluorescence assay. The virus-infected cells were collected by centrifugation to prepare cell particles and fixed in cacodylate sodium buffer containing 1% glutaraldehyde (0.2 M, pH 7.2). The fixed sample was dehydrated by acetone solution and embedded in epoxy resin. The sample was polymerized at 60°C for 3 days. The resin block was used to prepare ultrathin slices (50–70 nm thick). These sections were supported by copper mesh, negatively stained with uranyl acetate and lead citrate (electron microscopy) and observed with a JEM100SX transmission electron microscope (JEOL, Tokyo, Japan).

Biolayer interferometry

The Octet Red96 BLI platform (Sartorius BioAnalytical Instruments, Bohemia, USA) was utilized to measure the binding kinetics of three peptides to lipid A (L5399, Sigma, Shanghai, China). Peptides were diluted with 75 mM NaCl solution at concentrations of 100, 200, 400, and 600 nM. The amine reactive second-generation (AR2G) biosensors were processed with 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC, E1769, Sigma) and sulfo-N-hydroxysulfosuccinimide (s-NHS, 56485, Sigma). Subsequently, lipid A was loaded onto the activated AR2G biosensors at 30°C for 5 min, and both the processes of association and disassociation were monitored for 5 min to measure the kinetic constants. The equilibrium dissociation constant (K_D) was defined as the division of dissociation rate constant (K_{off}) and the association rate constant (K_{on}). In case of LPS, AR2G biosensors immobilized with 15 µg/mL LPS (L2880, Sigma) were incubated with 500 nM peptides for 2 min in 75 mM NaCl solution. The binding thickness was recorded and analyzed using ForteBio Data Analysis 7.0 software.

NPN uptake assay

MDRAB in mid-logarithmic phase was diluted to 1×10^8 CFU/mL in 10 mM sodium phosphate buffer (pH 7.4). Then 10 μ L of each peptide was mixed with 190 μ L of MDRAB at 37°C for 1 hour. Afterwards, 1-N-phenylmethylamine (NPN, 104043, Sigma, Shanghai, China) was added with a final concentration of 10 μ M. The fluorescence intensity was assessed with the wavelength ranging from 380 to 450 nm at 2 nm intervals, after excitation at 350 nm using a Tecan Infinite M1000 Pro microplate reader (Mannedorf, Zurich, Switzerland). Each experiment was conducted in duplicate and repeated three times.

Molecular dynamics simulations

The initial structures of peptides were built via AlphaFold2 single sequence version.⁵¹ Lipid bilayer models were constructed by CHARMM-GUI server along with placing parallel peptides approximately 2 nm above the membrane surface.⁵² Nine systems were built for three peptides versus three kinds of lipid bilayers (viral envelopes, Gram-negative bacterial inner membranes and outer membranes).^{39, 53} Ions of 150 mM NaCl were added to neutralize the system, and the calcium ions were used for LPS in outer leaflet of bacterial outer membranes. Detailed system settings are recorded in Table S7. The standard CHARMM-GUI equilibration protocols including NVT and NPT ensembles were adopted, keeping the temperature at 310K. All-atom MD simulations were performed using GROMACS 2021.2 with the CHARMM36 force field and the TIP3P water model for 100 ns, with a time step of 2 fs.^{54, 55} Three independent run was performed for each system. The visual molecular dynamics (VMD) software version 1.9.4 was used for visualization.

The distance between the center of mass (COM) of peptides and phosphorus atoms in top leaflets was calculated via *gmx distance*. Binding free energy was calculated using the molecular mechanics Poisson-Boltzmann surface area (MM/PBSA) method on average of extracted 20 frames from the last 20 ns. The binding free energy of a peptide-membrane complex is calculated as below:

$$\Delta G_{bind} = G_{complex} - (G_{peptide} + G_{membrane}) \quad (6)$$

$$G_{bind} = G_{polar} + G_{nonpolar} = (E_{elec} + G_{psolv}) + (E_{vdw} + G_{sasa}) \quad (7)$$

where E_{elec} is the Coulomb potential, E_{vdw} is the van der Waals energy, G_{psolv} and G_{sasa} are polar and nonpolar solvation free energy. Following previous study, the dielectric constant was set as 7 for phospholipids and the entropy term was neglected.⁵⁶

Data Availability

All the python codes of the AMP design model are available at <https://github.com/ruihan-dong/GAN-for-AMP-Design> and <https://github.com/ruihan-dong/AMPPredictor>. An online version of AMPPredictor can be accessed via <https://huggingface.co/spaces/ruihan-dong/AMPPredictor>.

Acknowledgements

We thank Prof. Chen Song from Peking University for helpful discussion in MD. We also thank Qiaozhen Meng, Yuqian Pu and Nan Song from Tianjin University. This work was supported by the National Key Research and Development Program of China (grant no. 2020YFA0908500), the

National Natural Science Foundation of China (grant no. 22007071, 31971127, 81971563, 82272330, 32100122, 62322215), Shanghai Sailing Program (21YF1457800), and Independent Research Project of State Key Laboratory of Trauma and Chemical Poisoning (SKLYQ202101).

Author Contributions

C.Z. and R.D. devised the project. C.Z., S.Y., and H.L. conceived the main conceptual ideas and were in charge of overall direction and planning. R.D. and F.G. developed the computational model. J.W., C.W., and G.Z. measured antimicrobial activities and performed animal experiments. X.W., R.L., Z.L., S.H., X.M., H.K., and J.L. carried out antiviral evaluations. Y.L. and P.Z. carried out anti-SARS-CoV-2 evaluations. C.Z. and R.D. analyzed the data and wrote the paper with input from all authors.

Conflict of Interest

There are no conflicts to declare.

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Editors

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Reviewer #1 (Public Review):

This manuscript presents a pipeline incorporating a deep generative model and peptide property predictors for the de novo design of peptide sequences with dual antimicrobial/antiviral functions. The authors synthesized and experimentally validated three peptides designed by the pipeline, demonstrating antimicrobial and antiviral activities, with one leading peptide exhibiting antimicrobial efficacy in animal models. However, the manuscript as it stands, has several major limitations on the computational side.

Major issues:

- (1) The choice of GAN as the generative model. There are multiple deep generative frameworks (e.g., language models, VAEs, and diffusion models), and GANs are known for their training difficulty and mode collapse. Could the authors elaborate on the specific rationale behind choosing GANs for this task?
- (2) The pipeline is supposed to generate peptides showing dual properties. Why were antiviral peptides not used to train the GAN? Would adding antiviral peptides into the training lead to a higher chance of getting antiviral generations?
- (3) For the antimicrobial peptide predictor, where were the contact maps of peptides sourced from?
- (4) Morgan fingerprint can be used to generate amino acid features. Would it be better to concatenate ESM features with amino acid-level fingerprints and use them as node features of GNN?
- (5) Although the number of labeled antiviral peptides may be limited, the input features (ESM embeddings) should be predictive enough when coupled with shallow neural networks. Have the authors tried simple GNNs on antiviral prediction and compared the prediction performance to those of existing tools?
- (6) Instead of using global alignment to get match scores, the authors should use local alignment.
- (7) How novel are the validated peptides? The authors should run a sequence alignment to get the most similar known AMP for each validated peptide, and analyze whether they are similar.
- (8) Only three peptides were synthesized and experimentally validated. This is too few and unacceptable in this field currently. The standard is to synthesize and characterize several dozens of peptides at the very least to have a robust study.

<https://doi.org/10.7554/eLife.97330.1.sa2>

Reviewer #2 (Public Review):

Summary:

This study marks a noteworthy advance in the targeted design of AMPs, leveraging a pioneering deep-learning framework to generate potent bifunctional peptides with specificity against both bacteria and viruses. The introduction of a GAN for generation and a GCN-based AMPredictor for MIC predictions is methodologically robust and a major stride in computational biology. Experimental validation in vitro and in animal models, notably with the highly potent P076 against a multidrug-resistant bacterium and P002's broad-spectrum viral inhibition, underpins the strength of their evidence. The findings are significant, showcasing not just promising therapeutic candidates, but also demonstrating a replicable means to rapidly develop new antimicrobials against the threat of drug-resistant pathogens.

Strengths:

The de novo AMP design framework combines a generative adversarial network (GAN) with an AMP predictor (AMPredictor), which is a novel approach in the field. The integration of deep generative models and graph-encoding activity regressors for discovering bifunctional AMPs is cutting-edge and addresses the need for new antimicrobial agents against drug-

resistant pathogens. The in vitro and in vivo experimental validations of the AMPs provide strong evidence to support the computational predictions. The successful inhibition of a spectrum of pathogens in vitro and in animal models gives credibility to the claims. The discovery of effective peptides, such as P076, which demonstrates potent bactericidal activity against multidrug-resistant *A. baumannii* with low cytotoxicity, is noteworthy. This could have far-reaching implications for addressing antibiotic resistance. The demonstrated activity of the peptides against both bacterial and viral pathogens suggests that the discovered AMPs have a wide therapeutic potential and could be effective against a range of pathogens.

<https://doi.org/10.7554/eLife.97330.1.sa1>

Reviewer #3 (Public Review):

Summary:

Dong et al. described a deep learning-based framework of antimicrobial (AMP) generator and regressor to design and rank de novo antimicrobial peptides (AMPs). For generated AMPs, they predicted their minimum inhibitory concentration (MIC) using a model that combines the Morgan fingerprint, contact map, and ESM language model. For their selected AMPs based on predicted MIC, they also use a combination of antiviral peptide (AVP) prediction models to select AMPs with potential antiviral activity. They experimentally validated 3 candidates for antimicrobial activity against *S. aureus*, *A. baumannii*, *E. coli*, and *P. aeruginosa*, and their toxicity on mouse blood and three human cell lines. The authors select their most promising AMP (P076) for in vivo experiments in *A. baumannii*-infected mice. They finally test the antiviral activity of their 3 AMPs against viruses.

Strengths:

- The development of de novo antimicrobial peptides (AMPs) with the novelty of being bifunctional (antimicrobial and antiviral activity).
- Novel, combined approach to AMP activity prediction from their amino acid sequence.

Weaknesses:

- I missed justification on why training AMPs without information of their antiviral activity would generate AMPs that could also have antiviral activity with such high frequency (32 out of 104).
- The justification for AMP predictor advantages over previous tools lacks rationale, comparison with previous tools (e.g., with the very successful AMP prediction approach described by Ma et al. 10.1038/s41587-022-01226-0), and proper referencing.
- Experimental validation of three de novo AMPs is a very low number compared to recent similar studies.
- I have concerns regarding the in vivo experiments including i) the short period of reported survival compared to recent studies (0.1038/s41587-022-01226-0, 10.1016/j.chom.2023.07.001, 0.1038/s41551-022-00991-2) and ii) although in Figure 2 f and g statistics have been provided, log scale y-axis would provide a better comparative representation of different conditions.
- I had difficulty reading the story because of the use of acronyms without referring to their full name for the first time, and incomplete annotation in figures and captions.

<https://doi.org/10.7554/eLife.97330.1.sa0>

Author response:

We thank the reviewers for their help and their suggestions to make this manuscript more rigorous. We would like to post provisional author responses when eLife publish the reviewed preprint, and the more detailed responses will be supplemented with the revised manuscript.

- *There are questions about choices made in the computational approach (architecture and type of generative model, training set).*

We will train a new generator model based on the current GAN architecture, but with 'hybrid' AMP/AVP training sets (Reviewer 1 and 3). Hence, we can directly compare the performances of two generators. Based on our preliminary data, providing GAN with more AVP sequences during training helped the designed peptides pass the AVP filter, at the cost of reducing the average AMPredictor scores. The new generator also elevated the diversity of designed sequences.

We also perturbed the detailed architecture of our deep learning models, including fully-connected graph edge encodings and different versions of ESM (e.g. esm1b_t33_650M_UR50S, esm2_t48_15B_UR50D, Reviewer 2). In the revised manuscript, we will report the effects of these modifications and suggest the overall construct of GCN and GAN are suitable for a light-weight sequence label model, as demonstrated in Author response table 1 and 2. For the generator, we suggest that using our approach, we may have reached a plateau for the GAN sampling (Author response table 3).

Author response table 1.

Results of AMPredictor with different graph edge encodings

Graph edge encoding	RMSE	MSE	Pearson	CI
Peptide Contact Map	0.5348	0.2860	0.7072	0.7294
<u>Fully-connected</u>	0.5319	0.2828	0.6946	0.7250

Author response table 2.

Results of AMPredictor with different ESM versions

ESM embedding	RMSE	MSE	Pearson	CI
esm1b_t33_650M_UR50S	0.5348	0.2860	0.7072	0.7294
esm2_t33_650M_UR50D	0.5425	0.2943	0.6944	0.7563
esm2_t36_3B_UR50D	0.5581	0.3115	0.6774	0.7494
esm2_t48_15B_UR50D	0.5622	0.3160	0.6503	0.7093

Author response table 3.

Evaluation of generated sequences with different sampling numbers

Sampling batch size	Predicted MICs (mean $\pm$ std)	Ratio of sequences with MIC < 5 μ M (mean $\pm$ std)
128	24.05 $\pm$ 1.13	0.193 $\pm$ 0.052
256	25.36 $\pm$ 0.52	0.160 $\pm$ 0.006
512	25.66 $\pm$ 1.51	0.188 $\pm$ 0.016
1024	24.11 $\pm$ 0.67	0.186 $\pm$ 0.004

- *There is an important concern about the small number of antimicrobial peptides tested, compared to other studies, and the origin of antiviral activities.*

We will address this concern by increasing the number of peptides tested in anti-microbial and anti-viral experiments. As reported in current version of our manuscript, the first generation of GAN generated 128 unique designs and the top 2% (3 designs) was tested experimentally. The second generation of GAN will produce ~1024 designs (1-2 weeks) and the top 2% (~ 20 new sequences) will be tested. We are in the process of synthesize (2-3 weeks) and MIC measurement (1 week). The overall size of tested sample will reach 20-30 sequences. We will focus on sequences with low similarity (< 30%) to any known AMPs, thus expanding the universe functional peptides. We estimated the collection of these new data in 6 weeks.

<https://doi.org/10.7554/eLife.97330.1.sa4>